

Rohit Kumar Gupta

Berlin, Germany | Willing to Relocate

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Profile

Software engineer with 3+ years of professional experience building production-grade APIs, microservices and ML-powered systems in Python. Strengths across REST API design (Django, Flask, FastAPI), database architecture, containerised deployment and CI/CD, combined with hands-on delivery of machine learning, NLP and MLOps solutions end-to-end. Currently completing a Master's in Artificial Intelligence in Berlin.

Technical Skills

- **Backend Frameworks:** Django, Flask, FastAPI, REST APIs, GraphQL, WebSocket, ORM (SQLAlchemy, Django ORM)
- **ML/DL Frameworks:** TensorFlow, PyTorch, scikit-learn, HuggingFace Transformers, XGBoost
- **NLP, GenAI & Vector Search:** BERT, Sentence-BERT, FAISS, Chroma, Embeddings, Semantic Search, LLM API integration (Groq), agent pipelines, RAG
- **Databases:** PostgreSQL, MySQL, SQLite, Query Optimization, Indexing, Migrations
- **MLOps & Deployment:** Docker, MLflow, DVC, Model Serving, CI/CD (GitHub Actions), Model Monitoring, Data Drift Detection
- **Data & Software Engineering:** Pandas, NumPy, ETL/ELT Pipelines, Feature Engineering, Microservices, System Design, TDD, PyTest, JWT/OAuth
- **Languages & Tools:** Python, C++, SQL, JavaScript/Node.js, Git, Linux, Bash

Professional Experience

Software Engineer

01.2024 – 05.2025

Ocean Cognizance Pvt. Ltd.

Mumbai, India

- Built and deployed production RESTful APIs (FastAPI, Flask) with JWT authentication for real-time ML model inference.
- Developed end-to-end ML pipelines – preprocessing, feature engineering, training, evaluation and deployment – including NLP-based document processing and predictive analytics.
- Designed database schemas (PostgreSQL, MySQL) with proper normalisation and indexing, improving query performance by 30%.
- Built microservices with Docker containerisation; implemented CI/CD (GitHub Actions) with automated testing (PyTest), reducing deployment/post-release defects by 25%.
- Architected ETL/ELT pipelines and automated ingestion workflows for batch and real-time processing, reducing manual data handling by 60%.

Software Developer

09.2022 – 12.2023

Sulzer Pumps India

Mumbai, India

- Developed real-time calibration application in C++ (MFC framework) with event-driven architecture and multithreading across 15+ modules.
- Implemented serial communication protocols (RS232/Modbus) with robust error handling for hardware integration.
- Led full software development lifecycle for a 7-month project deployed across 3 manufacturing sites in Ireland, reducing deployment errors by 30% via an automated build pipeline.

Software Developer Intern

06.2021 – 10.2021

Qikkle Solutions Pvt. Ltd.

Mumbai, India

- Contributed to full-stack development with focus on backend API integration and database connectivity (MySQL).
- Implemented automated testing procedures (resolving 15+ defects) and accelerated feature delivery by 20% through efficient Git workflows.

Projects

- **AI-IDP – Intelligent Document Processing Platform:** Full-stack document platform (FastAPI backend, React/TypeScript frontend) with OCR (Tesseract), BERT-based classification, sentence-transformer embeddings and FAISS semantic search; achieved 95% extraction accuracy with an 18% F1-score improvement over baseline; JWT auth, rate limiting and Celery/Redis background workers.
- **End-to-End ML System with Deployment:** MLOps pipeline with FastAPI/gRPC/batch serving, Docker and Kubernetes manifests, CI/CD automation, experiment tracking (MLflow), data versioning (DVC), model monitoring and drift detection.
- **ARIA Voice Assistant:** Real-time voice assistant built on a 4-agent LLM pipeline (Groq API) with strict JSON contracts, Node.js, WebSocket streaming and PostgreSQL; voice activity detection with barge-in.

Education

Master of Science in Artificial Intelligence

10.2025 – Present

IU International University of Applied Sciences

Berlin, Germany

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Advanced Neural Networks, AI System Design.

Bachelor of Engineering in Computer Science

06.2018 – 08.2022

University of Mumbai

Mumbai, India

- **Relevant Coursework:** Algorithms & Data Structures, Object-Oriented Programming, Statistics, Probability Theory, Database Management, Software Engineering.

Certifications & Languages

- Microsoft Azure Machine Learning Fundamentals – LinkedIn Learning; MLOps Essentials for Developers and AI Engineers – LinkedIn Learning; BCG GenAI Job Simulation – Forage
- **Languages:** English – Full Professional Proficiency (C1); German – A2; Hindi – Native